

PROPOSAL

HPC CENTRE OF EXCELLENCE

Phase 1 — Foundation, Training & Innovation Lab

Initial Investment: ₹5,00,000 | Implementation Timeline: 3–6 Months

Submitted to

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1. Executive Summary

The proposed Phase-1 HPC Centre of Excellence (CoE) is designed around the client's initial investment capacity of ₹5 lakh. Rather than attempting a full dedicated HPC cluster at this stage, Phase 1 establishes a functional HPC innovation lab that can be used immediately for training, experimentation and project development.

The solution combines one high-performance GPU workstation, Linux/RHEL-compatible configuration for 30 existing computers, controlled AWS cloud access, a structured HPC training programme and guided practical projects. This creates the foundation for future expansion into a multi-node HPC cluster, research programme and industry POC centre.

2. Vision

Build a production-oriented learning and innovation environment that develops practical capability in HPC, parallel computing and cloud infrastructure, while creating a clear path toward a larger institutional HPC CoE.

The original one-year CoE plan identifies infrastructure, talent development, research and industry engagement as the core pillars. Phase 1 focuses on establishing the infrastructure and talent foundation within the ₹5 lakh constraint.

3. Phase-1 Objectives

- Establish an operational HPC laboratory with GPU computing capability.
- Configure 30 existing computers with a common Linux/RHEL-compatible development environment.
- Introduce practical HPC technologies including MPI, OpenMP, CUDA and Python.
- Provide controlled AWS cloud access for CPU/GPU and cloud-HPC experimentation.
- Engage an experienced HPC trainer/mentor for practical delivery.
- Develop guided student/faculty projects with measurable benchmarks.
- Create the technical and operational foundation for future HPC cluster expansion.

4. Proposed Phase-1 Architecture

User / Student / Faculty → Linux Lab & GPU Workstation → HPC Software Stack → AWS Cloud → Project & Research Environment

Layer	Phase-1 Components
Compute	High-performance CPU + NVIDIA CUDA-capable GPU workstation
Lab	30 Linux/RHEL-compatible systems; SSH and common developer environment
Cloud	AWS compute, storage and optional GPU instances with controlled usage
HPC Software	Linux, GCC/G++, OpenMPI, OpenMP, Python, Git, CUDA where supported
Development	C/C++, Python, Jupyter/remote development, Git/GitHub workflow
Training	Linux, HPC fundamentals, parallel programming, GPU computing and cloud HPC
Projects	Benchmarking, optimization, GPU workloads and distributed-computing projects

5. Hardware & Infrastructure Scope

5.1 HPC GPU Workstation

- Modern high-core-count CPU
- 64–128 GB RAM target
- NVIDIA CUDA-capable GPU
- 1–2 TB NVMe SSD target
- Reliable motherboard, power supply and cooling
- Linux/RHEL-compatible configuration
- High-speed Ethernet where practical

The final component model and specification should be selected based on availability, warranty, compatibility and performance-per-rupee at procurement time.

5.2 30-Computer Linux Lab

The existing 30 computers will be configured as a consistent HPC learning/development environment. The deployment will include Linux/RHEL-compatible OS setup, users/access, SSH, compilers, Python, Git, MPI/OpenMP and other approved development tools. GPU/CUDA tooling will be installed only on systems whose hardware supports it.

5.3 AWS Cloud Environment

- Create/configure an AWS learning and experimentation environment.
- Provide controlled CPU/GPU compute access when local resources are insufficient.
- Introduce cloud storage, networking and basic cloud-HPC concepts.
- Use budgets, quotas and shutdown policies to control consumption.
- Demonstrate hybrid on-premise + cloud HPC workflows.

AWS consumption charges should be treated as usage-based operational costs and monitored separately from the one-time setup budget.

6. HPC Software & Technology Stack

Area	Technologies
Operating System	Linux / RHEL-compatible environment
Languages	C, C++, Python
Parallel Computing	OpenMP, MPI / OpenMPI
GPU Computing	NVIDIA CUDA where supported
Development	Git/GitHub, Jupyter, remote development
Cloud	AWS compute, storage and networking
Benchmarking	CPU/GPU workload and application performance measurement

7. HPC Training & Academy

Training will be practical and project-oriented. Participants should spend significant time executing workloads, measuring performance and troubleshooting real environments.

Module	Core Topics	Indicative Format
HPC Foundation	Linux, shell, CPU/memory/storage/networking, HPC concepts	20 hours
HPC Administrator	Cluster concepts, users, scheduler concepts, storage, monitoring, troubleshooting	40 hours
HPC Developer	OpenMP, MPI, CUDA, Python parallel programming, profiling	40 hours
Advanced GPU Engineer	Advanced GPU programming, multi-GPU workflows, performance tuning	40 hours
Cloud HPC	AWS compute, storage, networking, cloud workload execution and cost control	Workshop / practical

The original CoE plan defines Foundation, Administrator, Developer and Advanced GPU Engineer tracks and proposes certification after curriculum validation and industry review.

8. Practical Project Programme

Projects will be selected to move participants from basic parallel programming to GPU and cloud-based workloads.

- Parallel matrix multiplication using MPI/OpenMP
- OpenMP-based image processing
- CUDA acceleration of a computational workload
- Parallel sorting and benchmarking
- GPU-accelerated data processing workload
- CPU vs GPU performance comparison
- Cloud execution of selected HPC workloads
- Application profiling and performance optimization

Each project should have a mentor, measurable benchmark, Git repository, documentation and final demonstration.

9. HPC Trainer / Mentor

- Deliver structured classroom and lab sessions.
- Guide participants through Linux, MPI/OpenMP, CUDA and cloud-HPC exercises.
- Design and review practical assignments.
- Mentor project teams and troubleshoot technical issues.
- Introduce benchmarking and performance-analysis practices.
- Review final projects and identify candidates for advanced research or industry POCs.

10. Deliverables

Category	Key Deliverables
Infrastructure	1 GPU/HPC workstation; 30 Linux-configured systems; network/access setup
Software	Common HPC development stack; compilers; MPI/OpenMP; Python; Git; CUDA where supported
Cloud	AWS environment setup, access model and usage-control guidelines
Training	HPC Foundation + advanced practical modules; labs and assignments
Projects	5–10 guided project opportunities; benchmarking and documentation
CoE Enablement	Operating guidelines, curriculum, project pipeline and future expansion roadmap

11. Indicative Budget — ₹5 Lakh

Component	Indicative Allocation
High-performance GPU workstation	₹2,75,000 – ₹3,25,000
Networking / UPS / accessories	₹30,000 – ₹50,000
30-system Linux deployment	₹25,000 – ₹40,000
AWS initial usage / credits	₹25,000 – ₹40,000
HPC software/environment setup	₹15,000 – ₹25,000
HPC trainer / mentor	₹50,000 – ₹75,000
Project mentoring & documentation	₹20,000 – ₹30,000
Total	Target: within ₹5,00,000

Budget is indicative and subject to final hardware selection, taxes, vendor quotations, trainer engagement model and AWS consumption. Existing computers are assumed to be available; their purchase cost is not included.

12. Implementation Timeline

Period	Activities
Weeks 1–2	Requirements, lab assessment, hardware finalization, CoE governance and implementation plan
Weeks 3–5	GPU workstation procurement/setup; Linux image and software stack preparation
Weeks 5–7	Deploy Linux environment across 30 systems; networking, SSH and user/access configuration
Weeks 6–8	AWS environment, budgets/controls and cloud-HPC practical setup
Weeks 7–12	HPC training, hands-on labs, project selection and mentoring
Months 4–6	Advanced workshops, project completion, benchmarking, demos and Phase-2 roadmap

13. Success Metrics for Phase 1

- 1 operational HPC GPU workstation.
- 30 systems configured for Linux-based HPC learning.

- Core HPC software stack operational.
- Students/faculty completing practical HPC labs.
- At least 5 guided projects initiated, with a target of 5–10 depending on cohort size.
- Projects demonstrate measurable performance comparisons or optimization.
- AWS cloud-HPC workflows demonstrated under controlled spend.
- Documented roadmap for future multi-node HPC expansion.

14. Future Expansion Roadmap

Phase	Focus	Indicative Direction
Phase 1	Foundation	₹5 lakh: GPU workstation + Linux lab + AWS + training + projects
Phase 2	HPC Lab Expansion	Additional CPU/GPU nodes, shared storage and stronger networking
Phase 3	Dedicated HPC Cluster	Multi-node CPU/GPU cluster, high-speed interconnect and scalable storage
Phase 4	Full CoE	Research, industry POCs, consulting, advanced Quantum computing and managed HPC

Expansion should be driven by measured workloads, student/research demand, grants, partnerships and industry revenue rather than infrastructure size alone.

15. Commercial & Sustainability Model

- HPC training programmes
- Corporate HPC training
- HPC architecture and performance consulting
- Industry POCs and application optimization
- Managed HPC / cluster administration
- Sponsored research and government/academic grants

The long-term objective is to progressively reduce dependence on capital funding through recurring training, consulting, research and industry services.

16. Assumptions & Exclusions

- Client provides the 30 existing computers and suitable lab space.

- Electrical power, basic furniture and standard internet connectivity are available.
- RHEL subscription/support, if specifically required, will be finalized separately based on institutional licensing requirements.
- Hardware prices and AWS usage are variable; final procurement will follow vendor quotations.
- Large-scale production HPC cluster infrastructure is outside Phase-1 scope.
- Advanced paid software licences are excluded unless specifically approved.

17. Recommended Decision

Approve Phase 1 as a foundation investment of up to ₹5 lakh, with the objective of creating a working HPC lab that can immediately support training and project development while preserving a technically credible path to a larger HPC CoE.

“Start Lean. Train Practically. Build Projects. Benchmark Everything. Scale with Demand.”

18. Conclusion

Phase 1 converts the initial ₹5 lakh investment into a practical and demonstrable HPC capability rather than an undersized standalone cluster. By combining one strong GPU workstation, 30 Linux-enabled systems, AWS cloud access, expert HPC training and project mentoring, the Department of Computer Science can establish the essential foundations of a Centre of Excellence and generate evidence for future investment.

The proposed model is intentionally scalable: infrastructure can grow as real workloads, research requirements, partnerships and industry opportunities emerge.

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